Exercise 2.3 In the comparison shown in Figure 2.2, which method will perform best in the long run in terms of cumulative reward and probability of selecting the best action? How much better will it be? Express your answer quantitatively. □

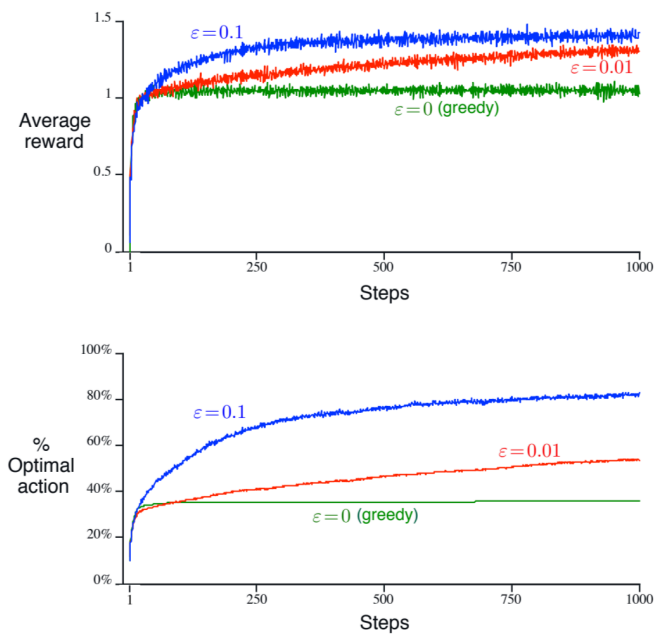

Figure 2.2: Average performance of ε-greedy action-value methods on the 10-armed testbed. These data are averages over 2000 runs with different bandit problems. All methods used sample averages as their action-value estimates.

$\varepsilon = 0.01$ will perform best.

OPTIMAL ACTION

$O(\varepsilon)$ - % of optimal action choices

$O(0.01) = (0.99 + 0.01 \cdot 0.1) \cdot 100 = (0.99 + 0.001) \cdot 100 = 99.1$

$O(0.1) = (0.9 + 0.1 \cdot 0.1) \cdot 100 = 91$

$O(0) \approx \frac{1}{3}$

AVERAGE REWARD

$q_* = 1.55$

ε	average reward
0	1
0.1	$0.91 \cdot 1.55 = 1.405$
0.01	$0.991 \cdot 1.55 = 1.53605$